

Day	Topic
Day 1	Go Basics - Setup & Fundamentals
Day 2	Control Flow & Functions
Day 3	Data Structures
Day 4	Interfaces & Error Handling
Day 5	Goroutines & Channels
Day 6	Packages & Modules
Day 7	Testing & Benchmarking

Day 8	Echo Framework - Setup & Basics
Day 9	Echo - Middleware & Validation
Day 10	Database Integration - PostgreSQL
Day 11	Authentication & Authorization
Day 12	Advanced Echo - File Upload & WebSockets
Day 13	API Documentation & Testing
Day 14	Deployment & Best Practices

Description
Install Go, understand workspace, basic syntax, variables, data types, operators
If/else, loops, switch statements, functions, multiple return values, defer
Arrays, slices, maps, structs, pointers, methods
Interface implementation, error handling patterns, custom errors, panic/recover
Concurrency basics, goroutines, channels, select, sync package, waitgroups
Creating packages, imports, go modules, dependency management, go.mod, go.sum
Writing unit tests, table-driven tests, benchmarks, test coverage, mocking

Install Echo, routing basics, handlers, context, middleware concept, request/response
Built-in middleware (logger, recover, CORS), custom middleware, request validation
Database/sql, GORM ORM basics, migrations, CRUD operations, connection pooling
JWT tokens, password hashing (bcrypt), protected routes, role-based access
File upload handling, static files, WebSocket connections, real-time features
Swagger/OpenAPI, API documentation, integration testing, mocking HTTP requests
Docker containerization, environment variables, logging, graceful shutdown, project structure

Documentation Links

<https://go.dev/doc/install>
<https://go.dev/tour/basics>
<https://gobyexample.com/variables>

<https://go.dev/tour/flowcontrol>
<https://gobyexample.com/if-else>
<https://gobyexample.com/functions>

<https://go.dev/tour/moretypes>
<https://gobyexample.com/slices>
<https://gobyexample.com/maps>
<https://gobyexample.com/structs>

<https://go.dev/tour/methods>
<https://gobyexample.com/interfaces>
<https://gobyexample.com/errors>
<https://go.dev/blog/error-handling-and-go>

<https://go.dev/tour/concurrency>
<https://gobyexample.com/goroutines>
<https://gobyexample.com/channels>
<https://gobyexample.com/channel-buffering>

<https://go.dev/doc/modules/managing-dependencies>
<https://go.dev/blog/using-go-modules>
<https://gobyexample.com/modules>

<https://go.dev/doc/tutorial/add-a-test>
<https://gobyexample.com/testing>
<https://pkg.go.dev/testing>

<https://echo.labstack.com/docs>
<https://echo.labstack.com/docs/quick-start>
<https://echo.labstack.com/docs/routing>

<https://echo.labstack.com/docs/middleware>
<https://echo.labstack.com/docs/middleware/logger>
<https://echo.labstack.com/docs/request>

<https://gorm.io/docs/>
<https://echo.labstack.com/docs/cookbook/crud>
<https://github.com/go-gorm/gorm>

<https://echo.labstack.com/docs/middleware/jwt>
<https://github.com/golang-jwt/jwt>
<https://pkg.go.dev/golang.org/x/crypto/bcrypt>

<https://echo.labstack.com/docs/cookbook/file-upload>
<https://echo.labstack.com/docs/cookbook/websocket>
<https://github.com/gorilla/websocket>

<https://github.com/swaggo/echo-swagger>
<https://github.com/swaggo/swag>
<https://echo.labstack.com/docs/testing>

<https://echo.labstack.com/docs/cookbook/graceful-shutdown>
https://go.dev/doc/effective_go
<https://github.com/golang-standards/project-layout>

Video Links

<https://www.youtube.com/watch?v=YS4e4q9oBaU>

(FreeCodeCamp Go Tutorial)

<https://www.youtube.com/watch?v=yyUHQlec83I> (Go in 100 Seconds)

<https://www.youtube.com/watch?v=YS4e4q9oBaU&t=3600s>

(Continue FreeCodeCamp)

<https://www.youtube.com/watch?v=LJvEljRBSDA> (Functions)

<https://www.youtube.com/watch?v=YS4e4q9oBaU&t=7200s>

<https://www.youtube.com/watch?v=p4LS3uGvHNI> (Slices Deep Dive)

<https://www.youtube.com/watch?v=YS4e4q9oBaU&t=10800s>

<https://www.youtube.com/watch?v=F4wdiWSW6kw> (Interfaces Explained)

<https://www.youtube.com/watch?v=YS4e4q9oBaU&t=14400s>

<https://www.youtube.com/watch?v=LvgVSSpwND8> (Goroutines)

<https://www.youtube.com/watch?v=Z1VhG7cf83M> (Go Modules)

<https://www.youtube.com/watch?v=aeF3I-zmPsY> (Packages)

https://www.youtube.com/watch?v=6AeqZvLj_Y8 (Testing Tutorial)

<https://www.youtube.com/watch?v=ndmB0bj7eyw> (Table Tests)

https://www.youtube.com/watch?v=_pww3NJuWnk (Echo Framework) https://www.youtube.com/watch?v=GD4KdOPQgCg (Echo REST API)
https://www.youtube.com/watch?v=J5lpwVuO9cs (Middleware) https://www.youtube.com/watch?v=qR0WnWL2o0Q (Validation)
https://www.youtube.com/watch?v=Yk5ZjKq4qDQ (GORM Tutorial) https://www.youtube.com/watch?v=QIVj_5wtPIw (Database)
https://www.youtube.com/watch?v=ma7rUS_vW9M (JWT Auth) https://www.youtube.com/watch?v=T8jYFbW-4Fs (Authentication)
https://www.youtube.com/watch?v=l9K-8wNAIbo (File Upload) https://www.youtube.com/watch?v=y9LU7nWxyyk (WebSockets)
https://www.youtube.com/watch?v=5tr1sgjEqU8 (Swagger) https://www.youtube.com/watch?v=G5J1EQkVBZ8 (API Testing)
https://www.youtube.com/watch?v=oNLkkqvomg (Docker Go) https://www.youtube.com/watch?v=XCsL89YtqCs (Production)

Practice Tasks	Duration (hrs)
Install Go, setup GOPATH, write hello world, practice variable declarations, type conversions	3-4
Write calculator functions, implement fizzbuzz, create temperature converter with multiple returns	3-4
Create student record system using structs, implement slice operations, build phonebook with maps	4-5
Implement Shape interface with multiple types, create file reader with proper error handling	4-5
Build concurrent URL checker, implement worker pool pattern, create pipeline with channels	5-6
Create multi-package project, initialize go module, add external dependencies, build CLI tool	3-4
Write tests for previous projects, implement table-driven tests, create benchmarks for functions	4-5

Create first Echo app, implement basic routes (GET, POST, PUT, DELETE), test with Postman	4-5
Add logger middleware, implement custom auth middleware, add request validation with validator.v10	4-5
Setup PostgreSQL, connect with GORM, create models, implement CRUD API endpoints	5-6
Implement user registration/login, JWT token generation, protect routes, add role-based middleware	5-6
Build file upload endpoint with validation, create image server, implement basic chat with WebSocket	5-6
Add Swagger docs to API, write integration tests, implement httptest for handlers	4-5
Create Dockerfile, setup env config, implement graceful shutdown, organize project structure, deploy demo	5-6

Resource Category	Resource Name
Official Docs	Go Official Documentation
Official Docs	Echo Framework Docs
Official Docs	Go Tour
Books	The Go Programming Language
Books	Learning Go (O'Reilly)
Practice	Go by Example
Practice	Exercism Go Track
Practice	LeetCode Go Problems
Community	r/golang
Community	Gophers Slack
Tools	GORM
Tools	golang-jwt
Tools	validator
Tools	Air
Tools	golangci-lint
Videos	FreeCodeCamp Go Course
Videos	TechWorld with Nana - Go
Videos	Anthony GG
Projects	Build a REST API

Link
https://go.dev/doc/
https://echo.labstack.com/docs
https://go.dev/tour/
https://www.gopl.io/
https://www.oreilly.com/library/view/learning-go/9781492077206/
https://gobyexample.com/
https://exercism.org/tracks/go
https://leetcode.com/
https://reddit.com/r/golang
https://gophers.slack.com/
https://gorm.io/
https://github.com/golang-jwt/jwt
https://github.com/go-playground/validator
https://github.com/cosmtrek/air
https://golangci-lint.run/
https://www.youtube.com/watch?v=YS4e4q9oBaU
https://www.youtube.com/watch?v=yyUHQlec83I
https://www.youtube.com/@anthonygg_
https://github.com/toptal/go-rest-api-example

Description
Complete Go language documentation
Official Echo framework documentation
Interactive Go tutorial
Comprehensive Go book by Donovan & Kernighan
Modern Go practices and idioms
Hands-on examples of Go concepts
Practice exercises with mentorship
Algorithm practice in Go
Go community on Reddit
Official Go community Slack
Go ORM library
JWT implementation for Go
Struct validation package
Live reload for Go apps
Fast Go linters runner
7-hour comprehensive Go course
Go tutorial series
Advanced Go concepts channel
Example REST API project

Project	Description
TODO API	RESTful API for task management with CRUD operations
Blog Platform API	API for blog posts with user auth, comments, tags
E-commerce Backend	Product catalog, cart, orders, payment integration
Real-time Chat	WebSocket-based chat with rooms and user presence
URL Shortener	Shorten URLs with analytics and custom aliases
File Sharing Service	Upload, share files with expiration and access control
Social Media API	Posts, follows, likes, feed generation
Microservices Demo	User service, product service with API gateway

Technologies	Difficulty
Echo, GORM, PostgreSQL, JWT	Beginner
Echo, GORM, JWT, File Upload	Intermediate
Echo, PostgreSQL, Redis, Stripe API	Advanced
Echo, WebSocket, Redis Pub/Sub	Intermediate
Echo, Redis, PostgreSQL	Beginner
Echo, S3/MinIO, JWT	Intermediate
Echo, PostgreSQL, Redis Cache	Advanced
Echo, gRPC, Docker, Kubernetes	Advanced